

EE115B *Digital Circuits*

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- 74xxyy series chip
- IC Package: pin identification, DIP, SOIC
- Datasheet:

Maximum ratings: V_{CC} V_{in}

Recommended operating conditions: V_{CC} V_{in}

DC characteristics: threshold voltages

AC characteristics: propagation time, transition time

1. $A + 0 = A$

2. $A + 1 = 1$

3. $A \cdot 0 = 0$

4. $A \cdot 1 = A$

5. $A + A = A$

6. $A + \bar{A} = 1$

7. $A \cdot A = A$

8. $A \cdot \bar{A} = 0$

9. $\bar{\bar{A}} = A$

10. $A + AB = A$

11. $A + \bar{A}B = A + B$ *

12. $(A + B)(A + C) = A + BC$ *



Advanced Boolean Theorems

	Function	Dual
Commutation	$A + B = B + A$	$AB = BA$
Association	$A + (B + C) = (A + B) + C$	$A(BC) = (AB)C$
Distribution	$A + BC = (A + B)(A + C)$	$A(B + C) = AB + AC$
Absorption	$A + AB = A$	$A(A + B) = A$
De Morgan	$\overline{A + B} = \bar{A} \cdot \bar{B}$	$\overline{AB} = \bar{A} + \bar{B}$
Consensus	$AC + B\bar{C} =$ $AB + AC + B\bar{C}$	$(A + C)(B + \bar{C}) =$ $(A + B)(A + C)(B + \bar{C})$



Consensus theorem

- The consensus theorem can be stated as follows:

$$XY + X'Z + YZ = XY + X'Z$$

- **Proof**

$$\begin{aligned} XY + X'Z + YZ &\rightarrow XY + X'Z + (X + X')YZ \\ &\rightarrow XY + X'Z + XYZ + X'YZ \\ &\rightarrow (XY + XYZ) + (X'Z + X'YZ) \\ &\rightarrow XY(1 + Z) + X'Z(1 + Y) \\ &\rightarrow XY + X'Z \end{aligned}$$

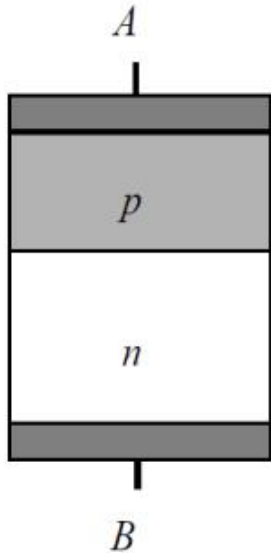


Low-level Gates

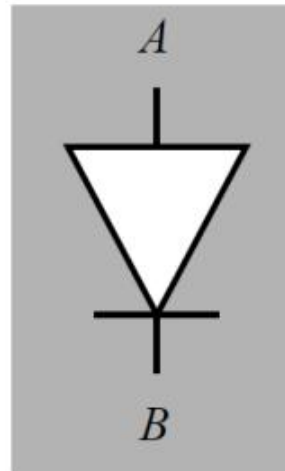
Hardware Implementation

Floyd Book, Chapter 15

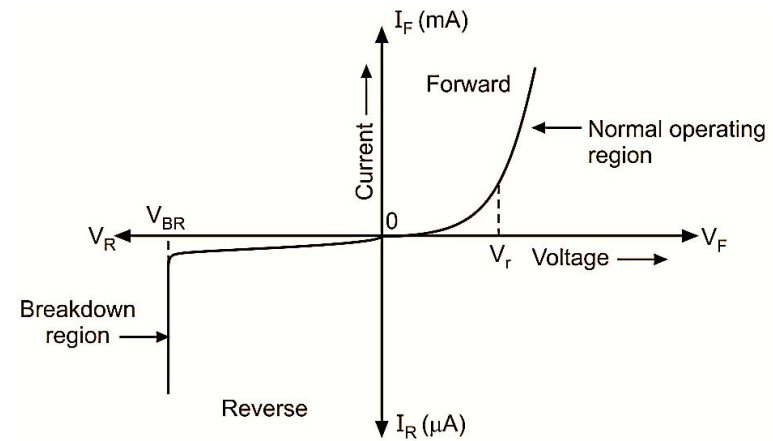
PN Junction

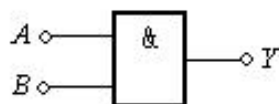
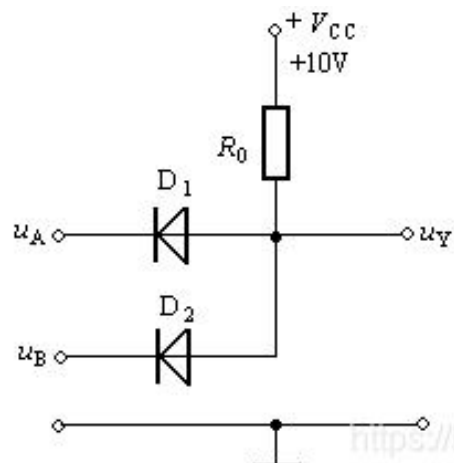


Schematics

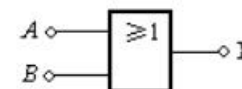
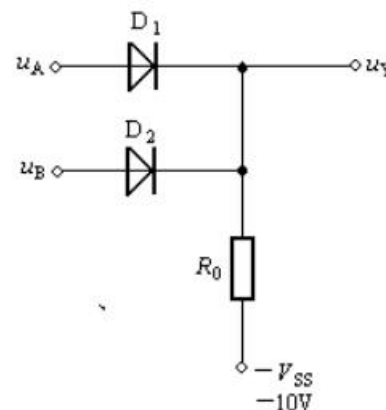


Circuit symbol





AND Logic



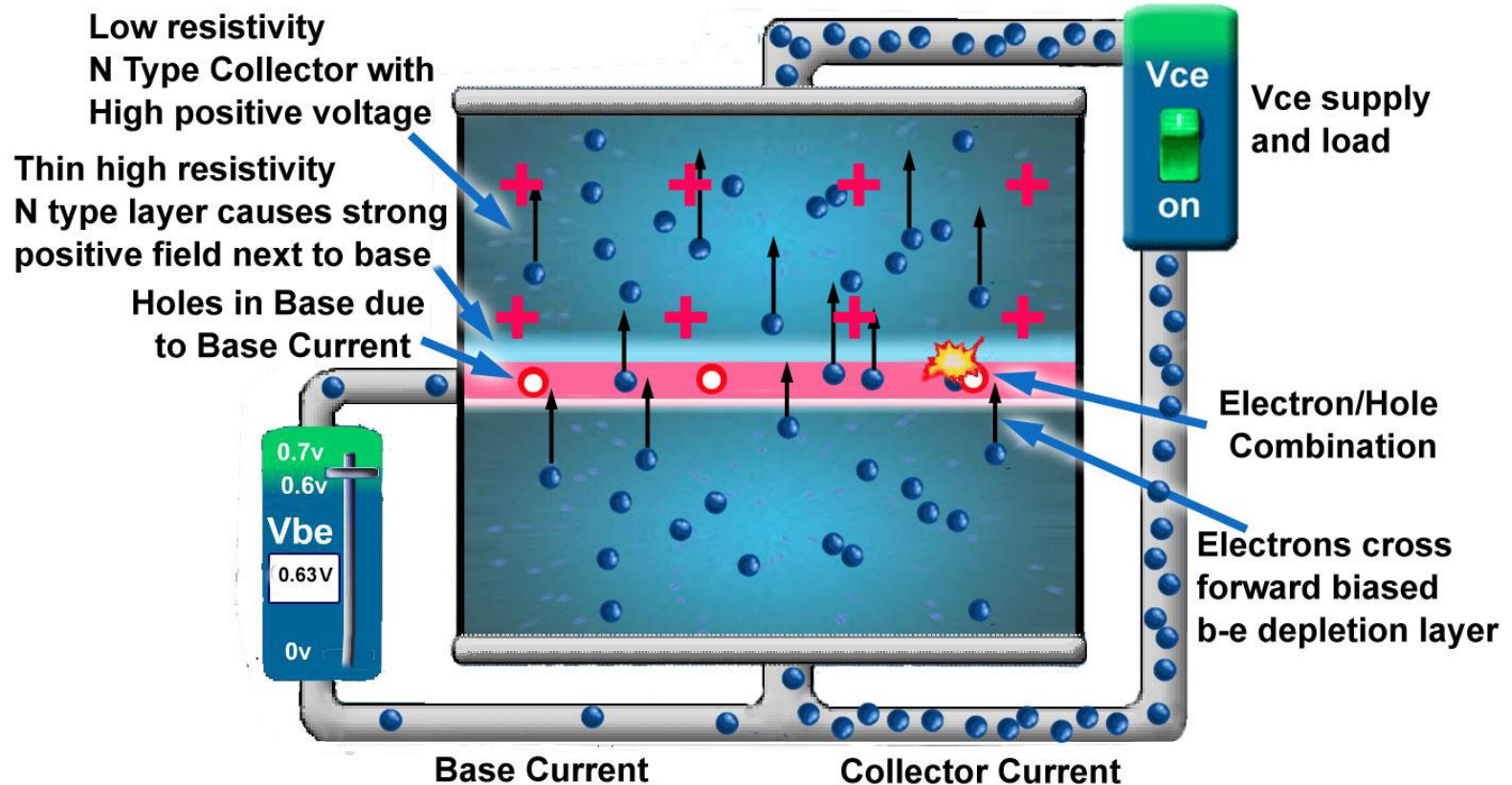
OR Logic

$-10V = 0$

$0V = 1$

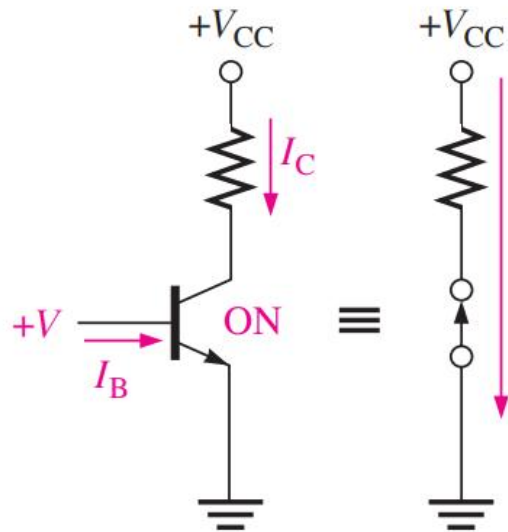


How BJT works

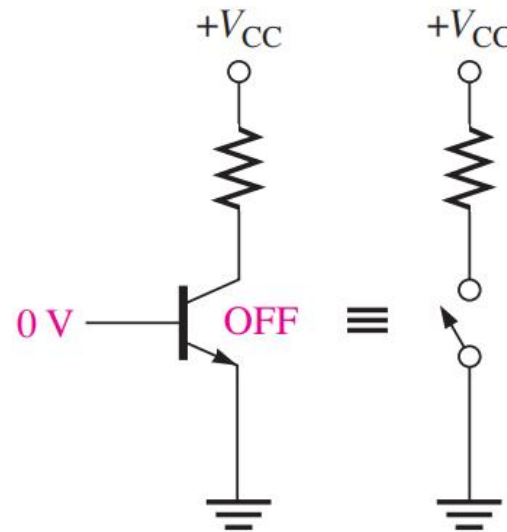


https://www.learnabout-electronics.org/Semiconductors/bjt_03.php

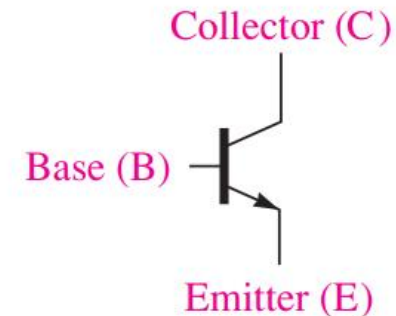
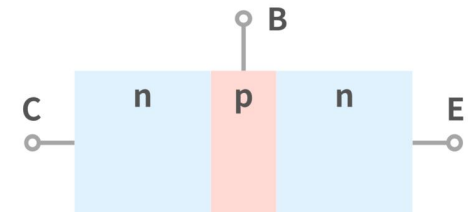
BJT (NPN)



(a) Saturated (ON) transistor and ideal switch equivalent



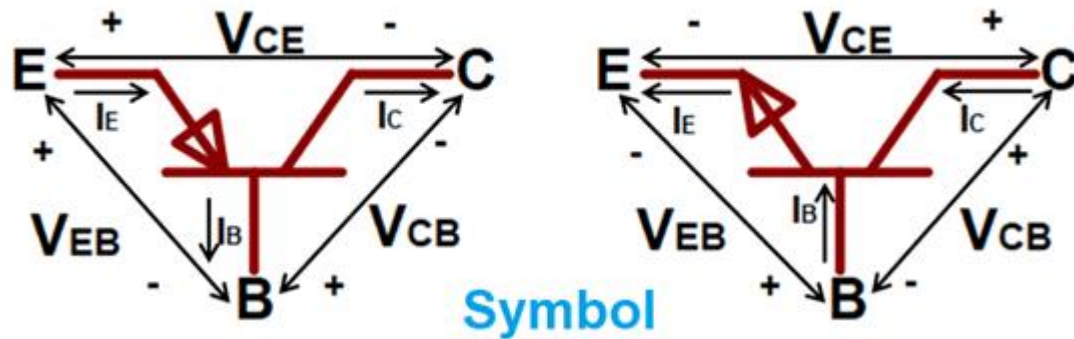
(b) OFF transistor and ideal switch equivalent



$V_{BE} > 0.7$ ON
 $V_{BE} < 0.7$ OFF

FIGURE 15-25 The symbol for a BJT.

BJT: NPN vs PNP



BJT Inverter

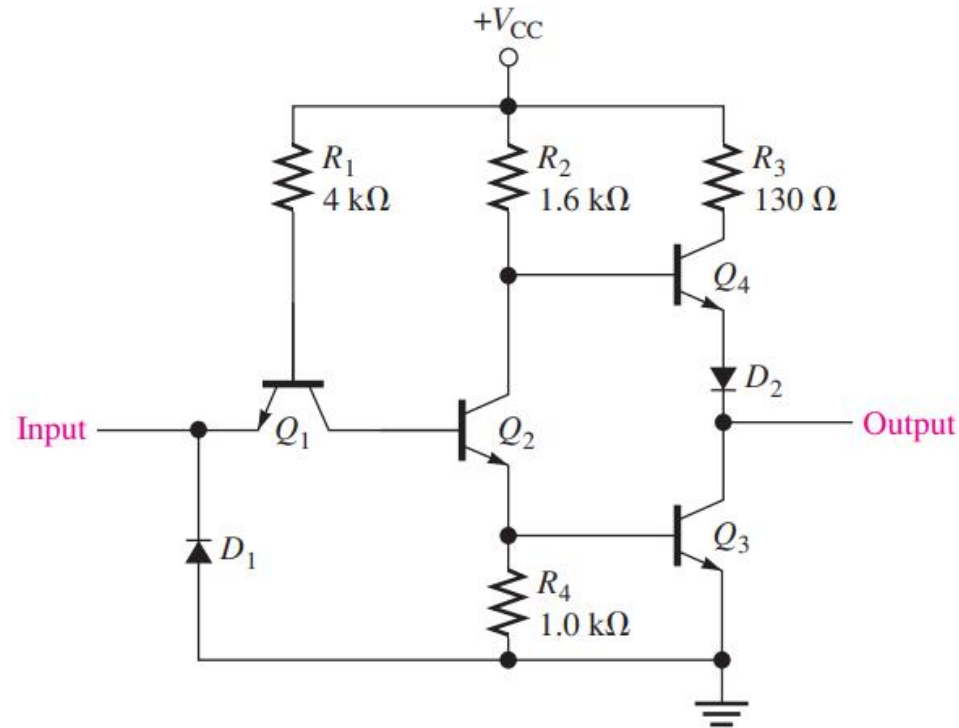
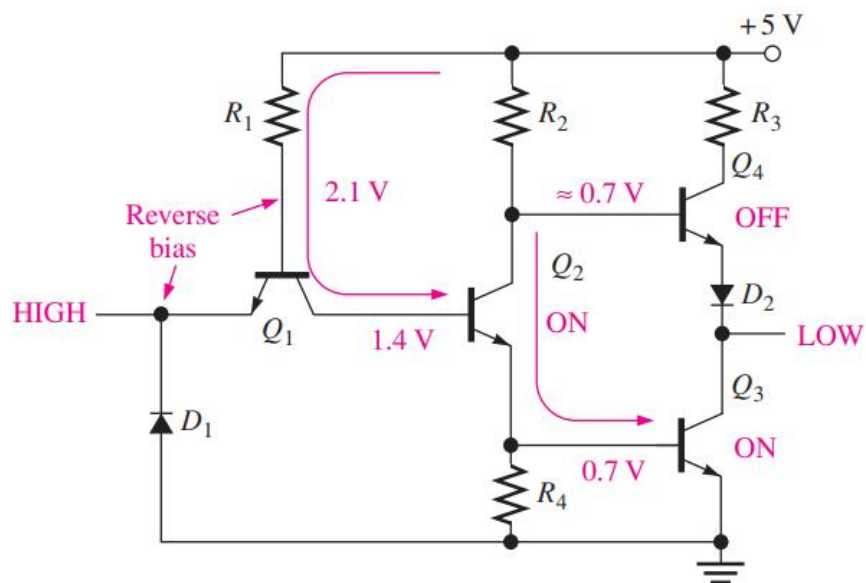
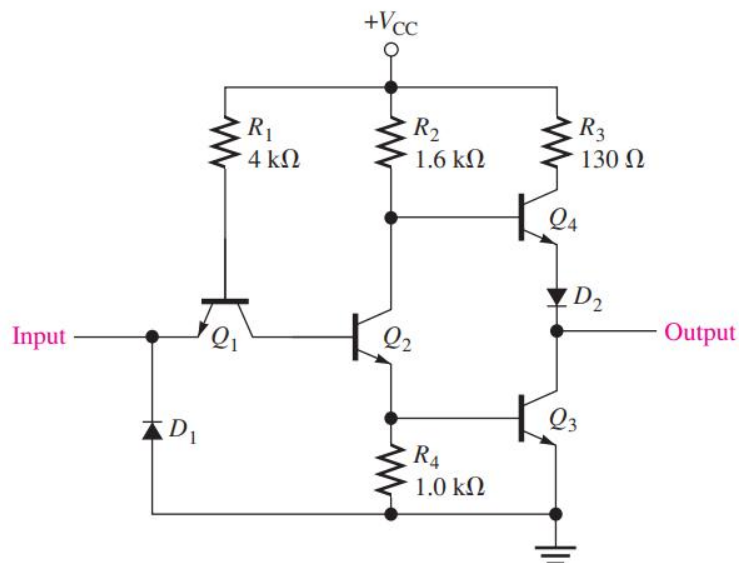
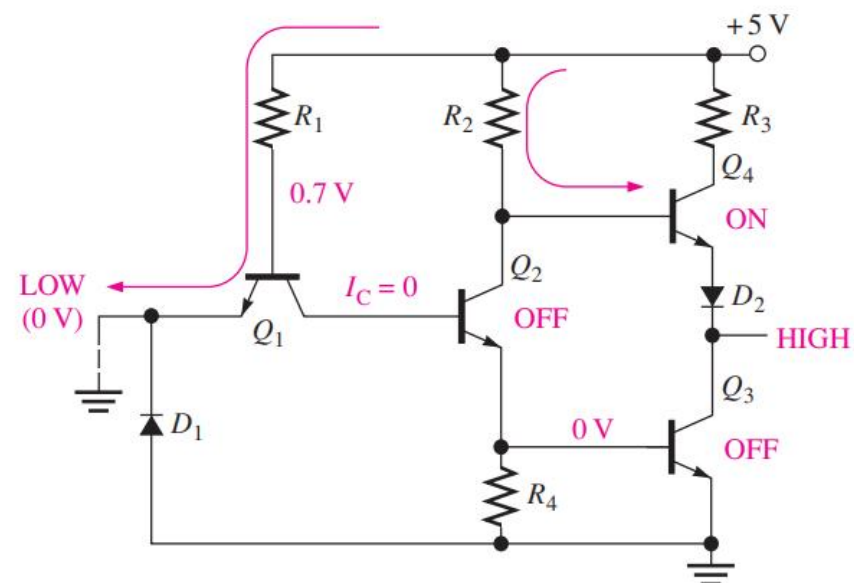


FIGURE 15-27 A standard TTL inverter circuit.



(a)



(b)

FIGURE 15-28 Operation of a TTL inverter.

TTL NAND Gate

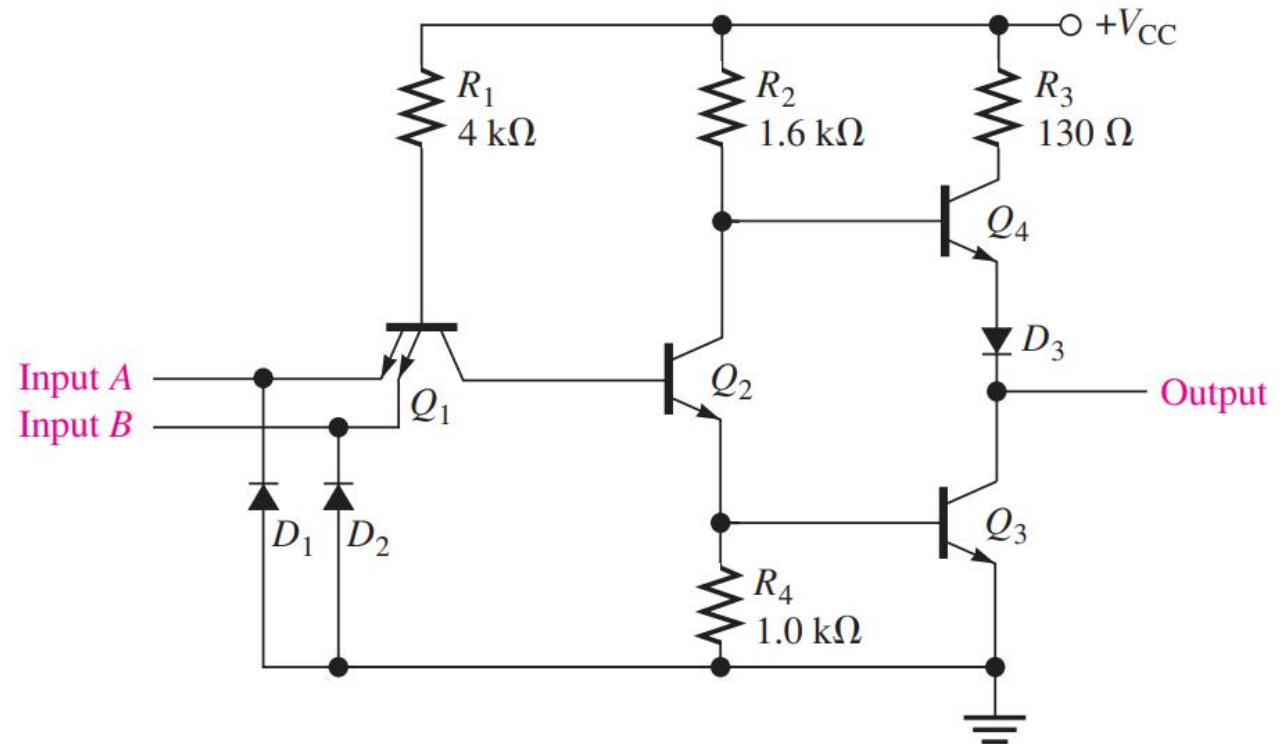
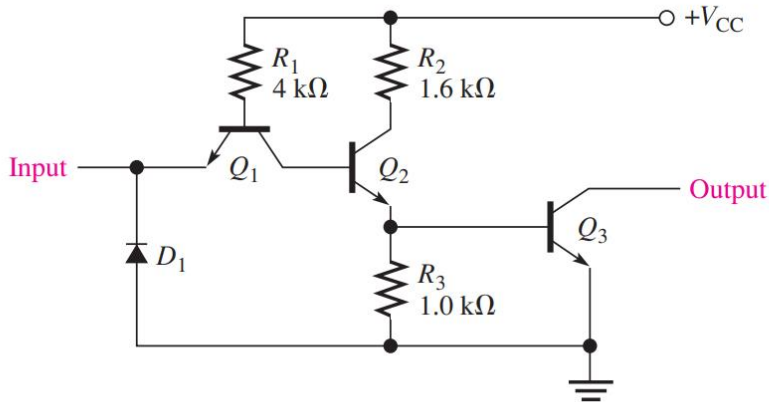


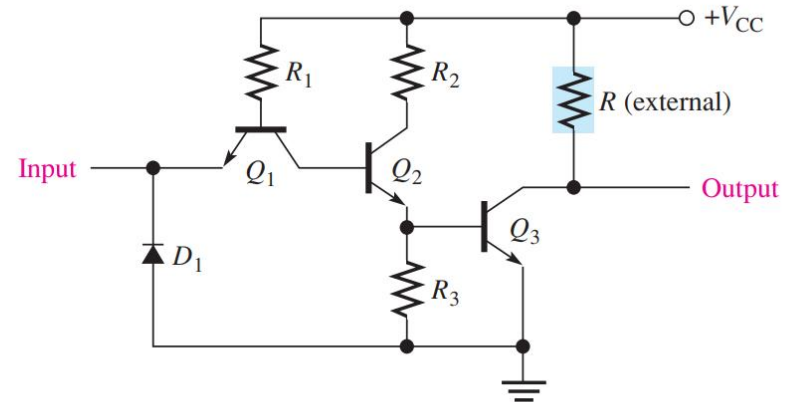
FIGURE 15-29 A TTL NAND gate circuit.



Open-Collector Gates



(a) Open-collector inverter circuit



(b) With external pull-up resistor

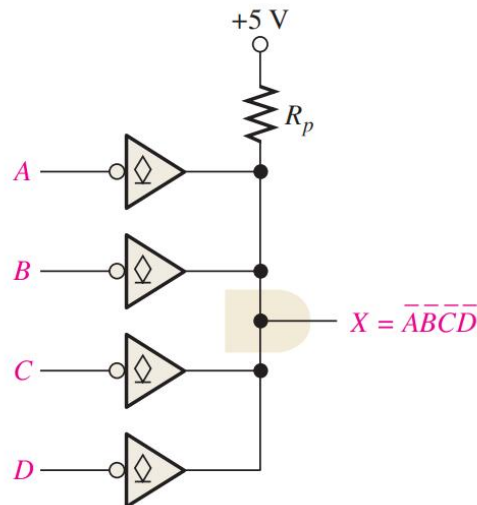
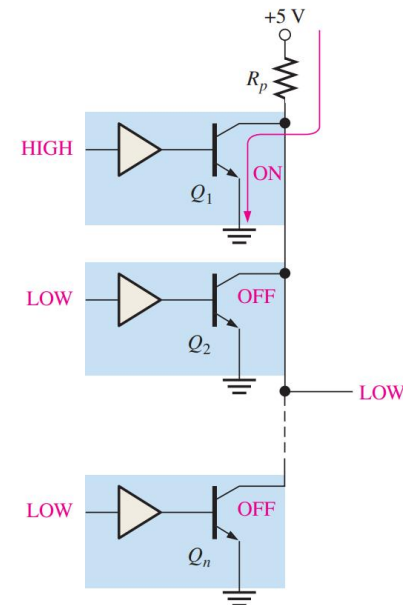
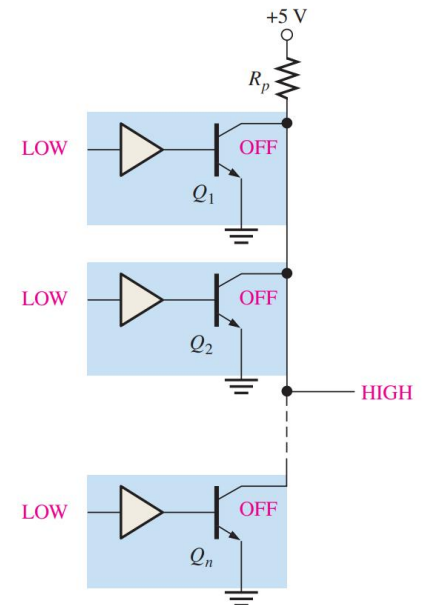


FIGURE 15-37 A wired-AND configuration of four inverters.



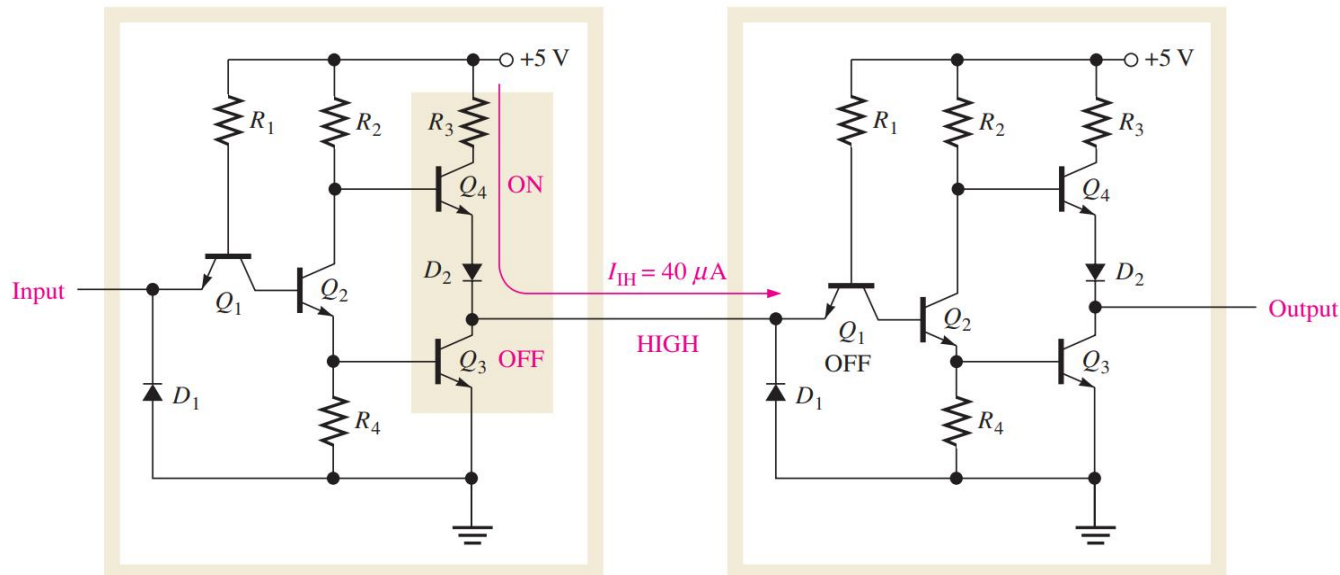
(a) When one or more output transistors are on, the output is LOW.



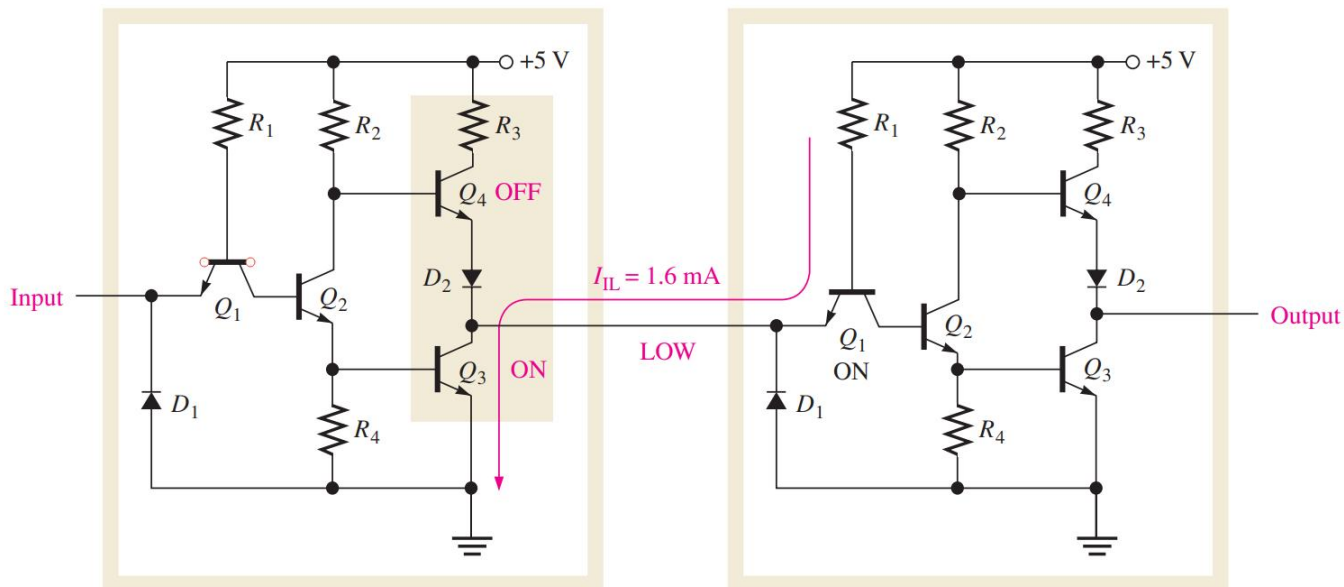
(b) When all output transistors are off, the output is HIGH.

Source and Sink Current

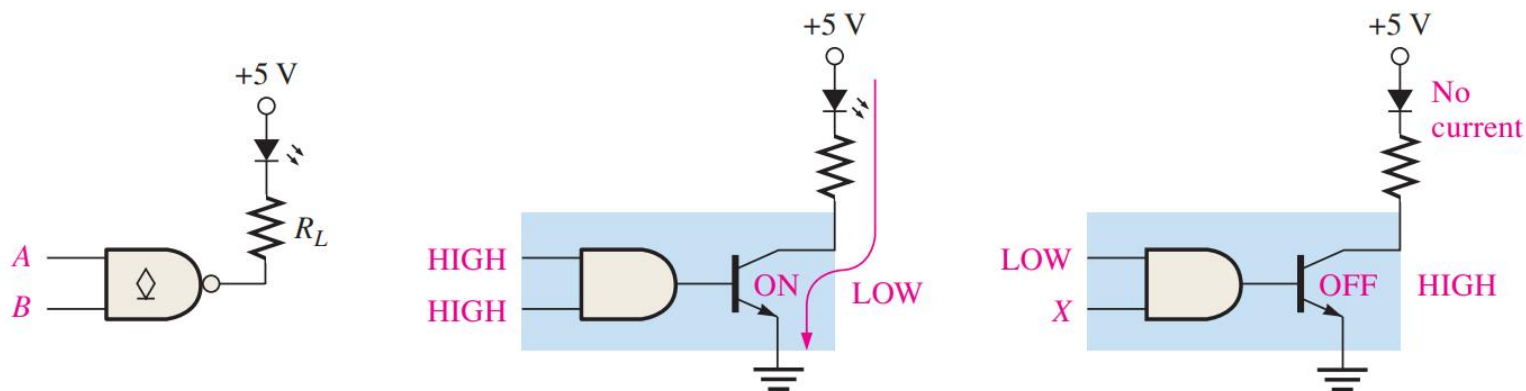
Source



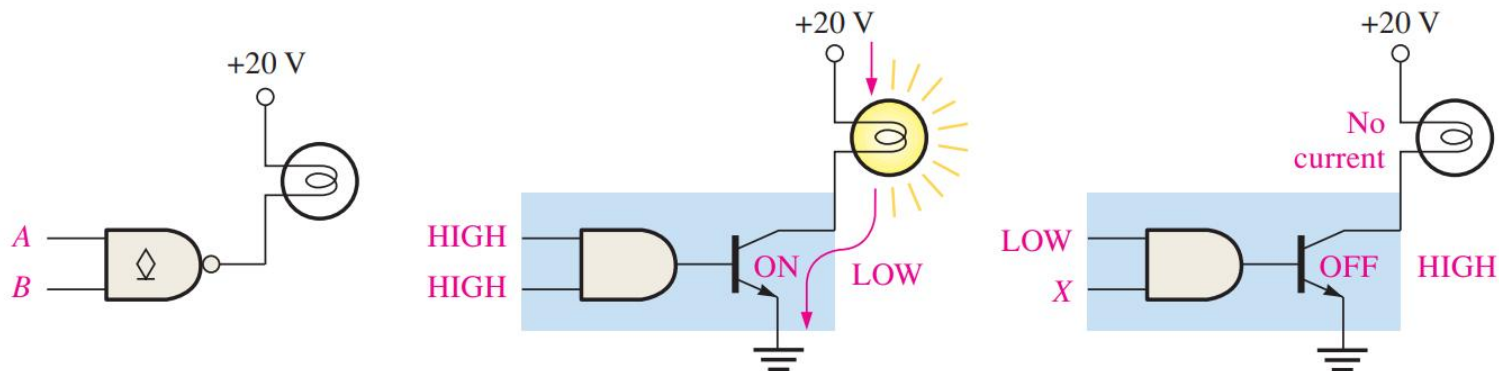
Sink



Source and Sink Current



(a) Driving an LED



(b) Driving a low-current lamp

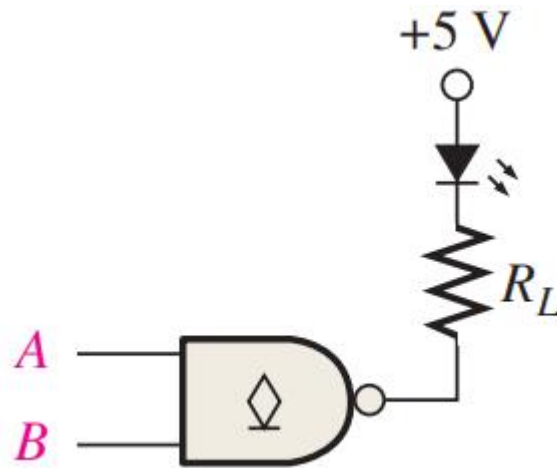
FIGURE 15-42 Some applications of open-collector drivers.



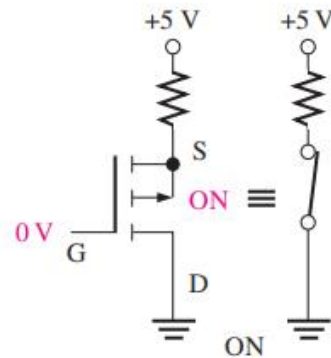
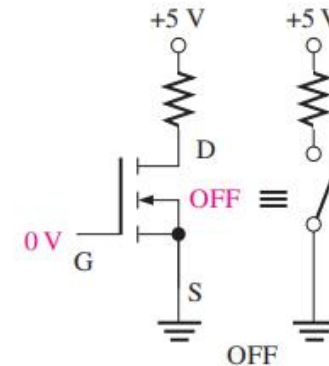
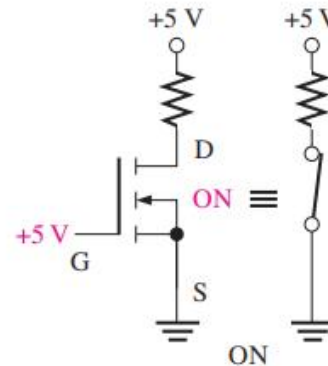
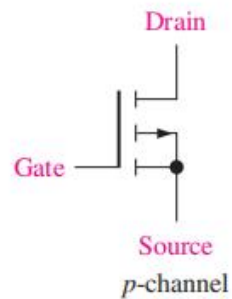
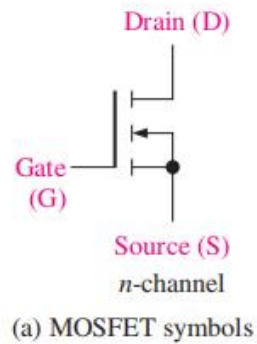
Source and Sink Current

Determine the value of the limiting resistor, R_L , in the open-collector circuit:

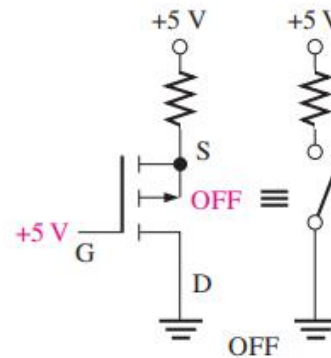
If the LED current is to be 20 mA. Assume a 1.5 V drop across the LED when it is forward-biased and a LOW-state output voltage of 0.1 V at the output of the gate.

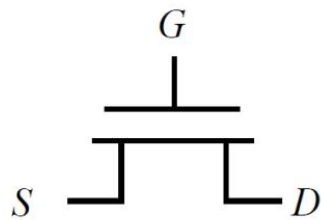
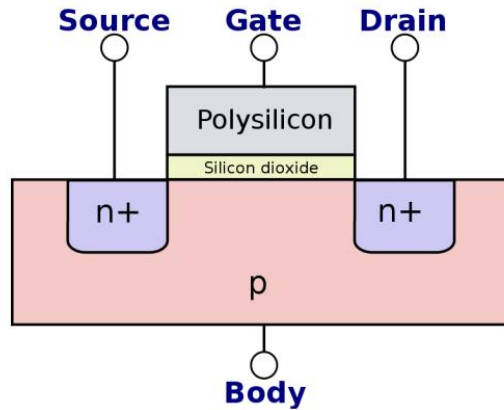


MOSFET

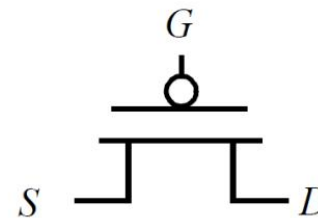
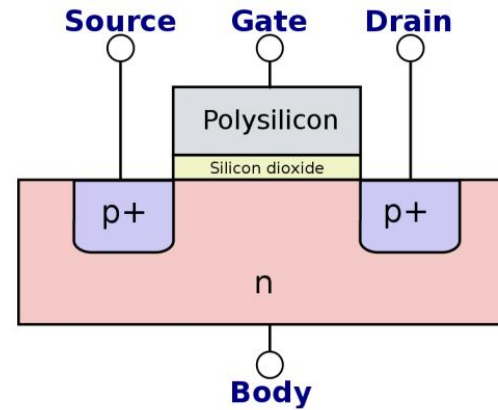


(c) *p*-channel switch





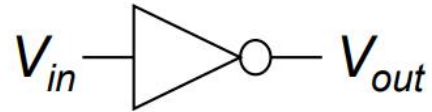
- NMOS
- Body connected to GND



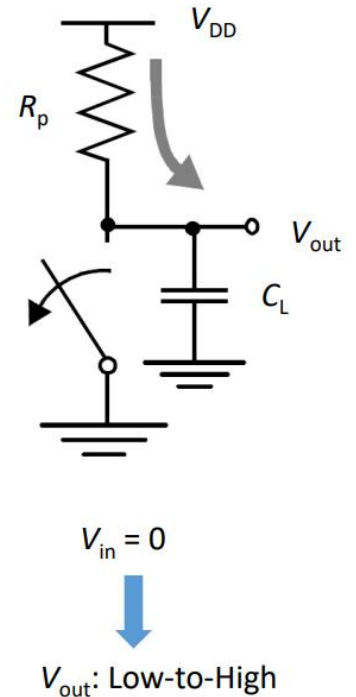
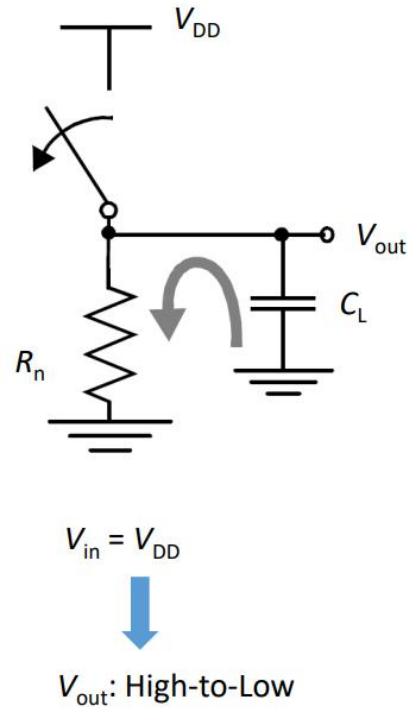
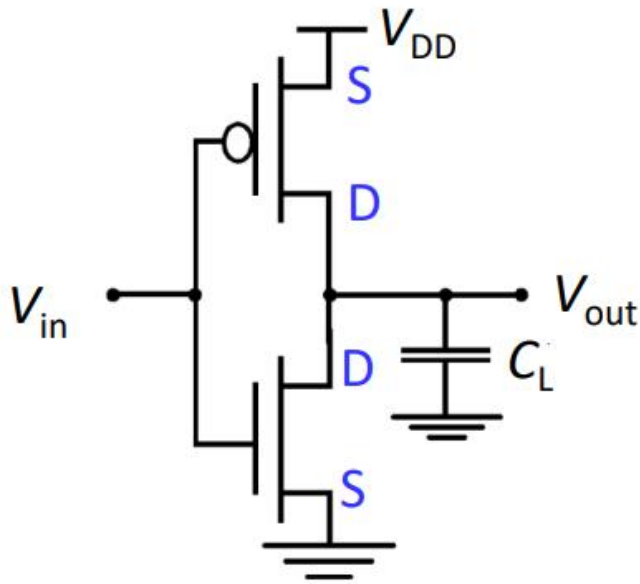
- PMOS
- Body connected to V_{DD}

The CMOS Inverter

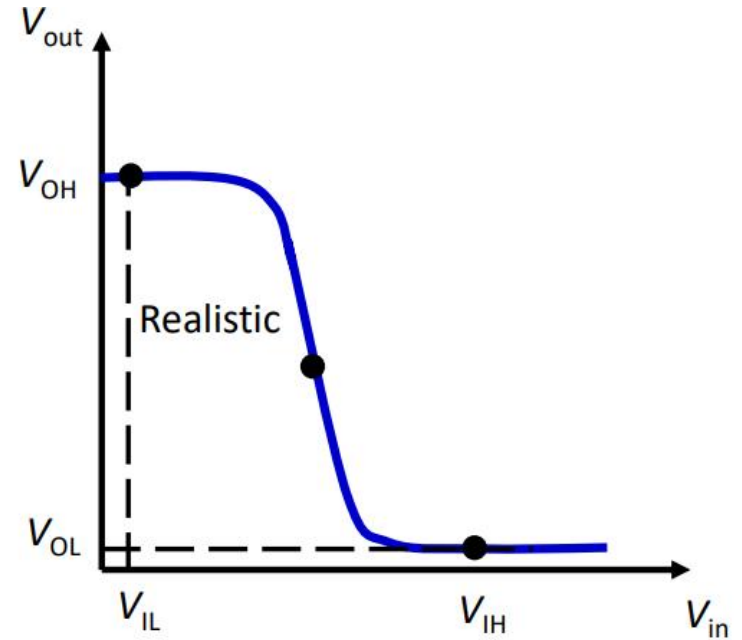
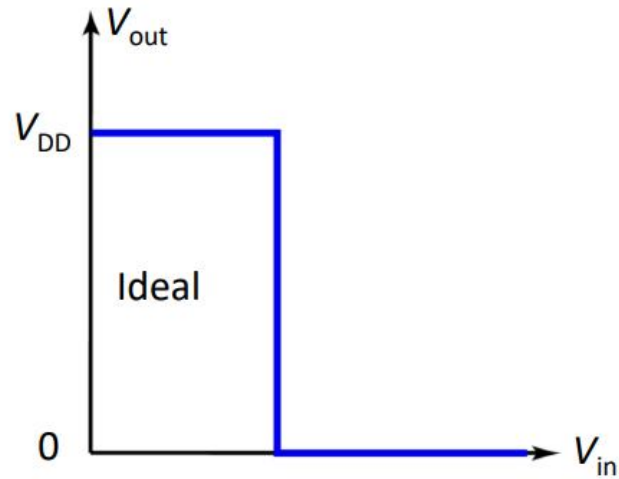
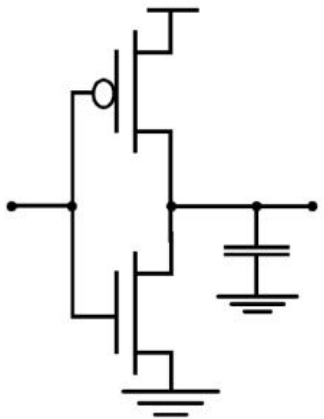
Symbol



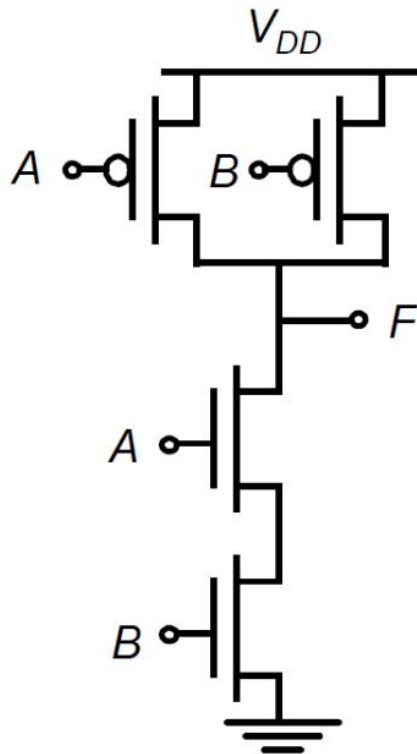
Schematic



Inverter Voltage-Transfer Characteristic (VTC)



NAND gate

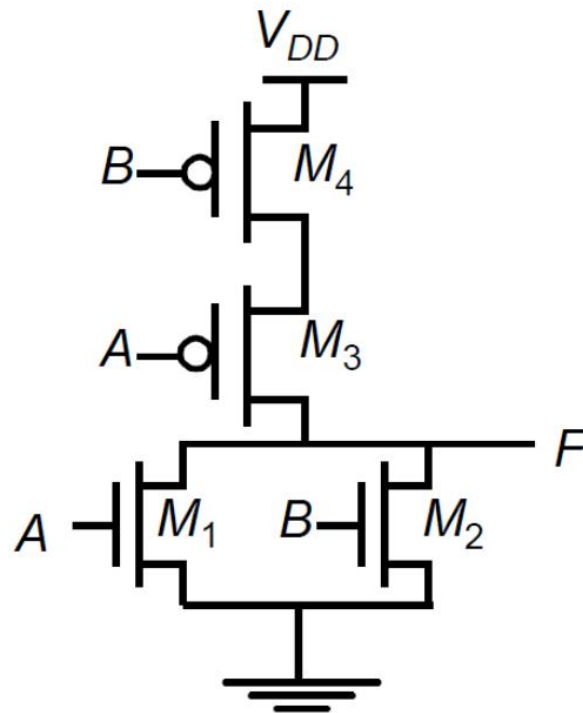


Circuit

Truth table

Inputs		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

NOR gate



Circuit

Truth table

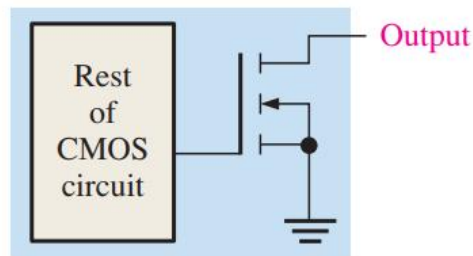
Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	1
0	1	0
1	0	0
1	1	0



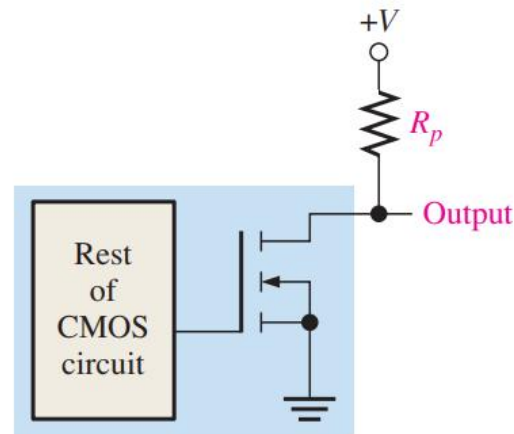
Open-Drain Gates

Open-Drain Gates

The term *open-drain* means that the drain terminal of the output transistor is unconnected and must be connected externally to V_{DD} through a load. An open-drain gate is the CMOS counterpart of an open-collector TTL gate (discussed in Section 15–3). An open-drain output circuit is a single n -channel MOSFET as shown in Figure 15–21(a). An external **pull-up resistor** must be used, as shown in part (b), to produce a HIGH output state. Also, open-drain outputs can be connected in a wired-AND configuration, a concept that is discussed in Section 15–4 in relation to TTL.



(a) Unconnected output

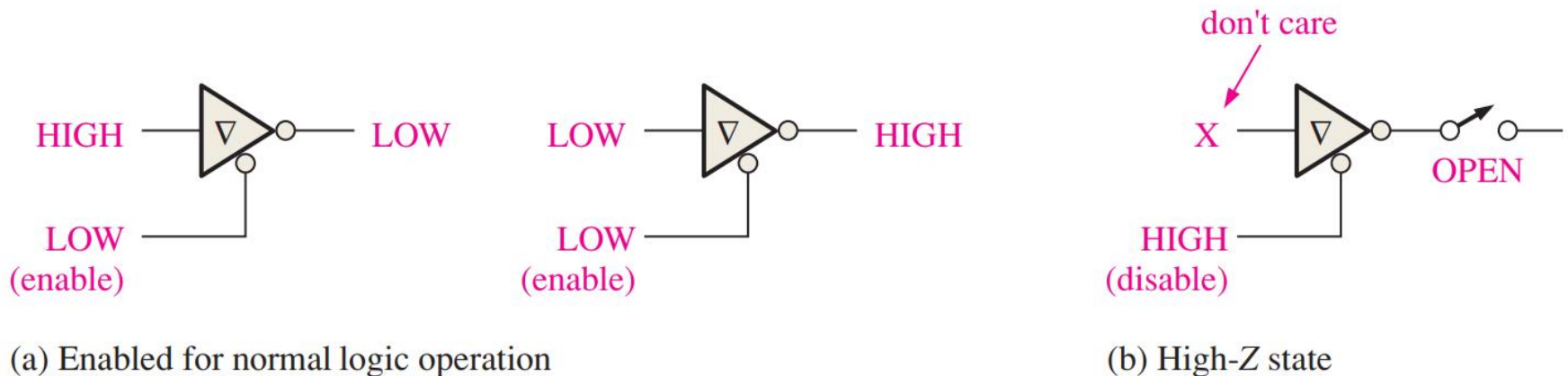


(b) With pull-up resistor

Tristate Gates

Tri-state CMOS Gates

Tri-state outputs are available in both CMOS and TTL logic. The **tri-state** output combines the advantages of the totem-pole and open-collector circuits. As you recall, the three output states are HIGH, LOW, and high-impedance (**high-Z**). When selected for normal logic-level operation, as determined by the state of the enable input, a tri-state circuit operates in the same way as a regular gate. When a tri-state circuit is selected for high-Z operation, the output is effectively disconnected from the rest of the circuit by the internal circuitry. Figure 15–22 illustrates the operation of a tri-state circuit. The inverted triangle (∇) designates a tri-state output.



Loading and Fan-Out

When the output of a logic gate is connected to one or more inputs of other gates, a load on the driving gate is created, as shown in Figure 15–10.

There is a limit to the number of load gate inputs that a given gate can drive. This limit is called the fan-out of the gate.

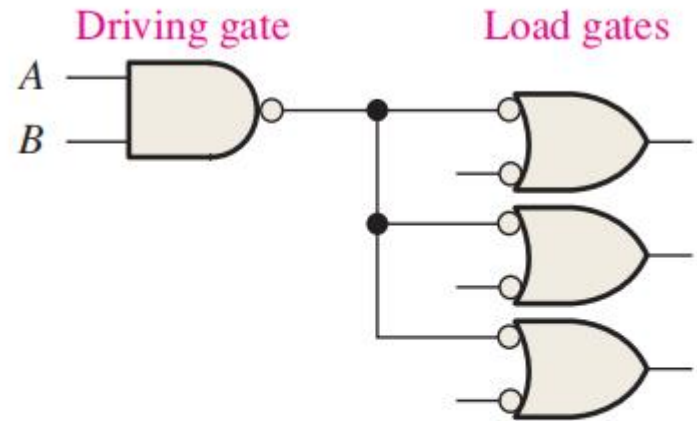


FIGURE 15–10 Loading a gate output with gate inputs.



Loading and Fan-Out

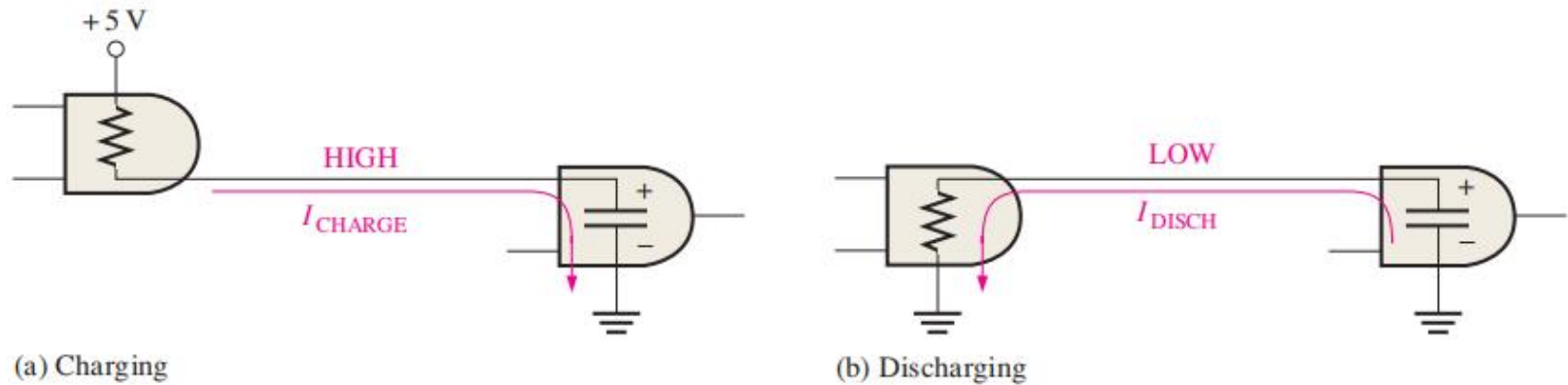


FIGURE 15-11 Capacitive loading of a CMOS gate.

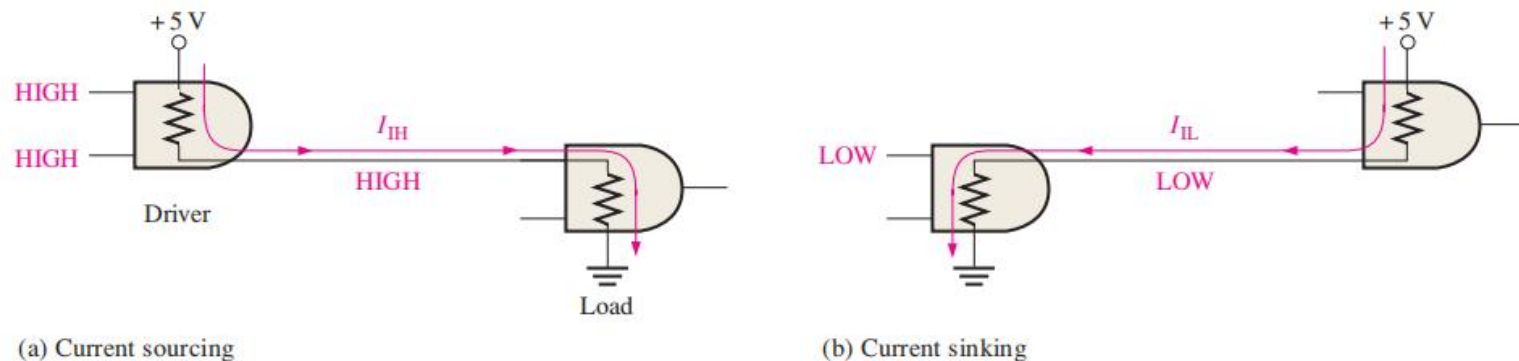


FIGURE 15-12 Basic illustration of current sourcing and current sinking in logic gates.

Time Specifications

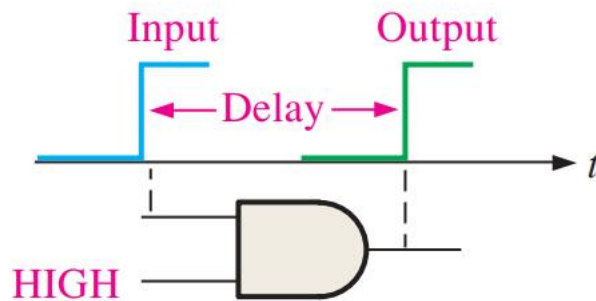


FIGURE 15-8 A basic illustration of propagation delay time.

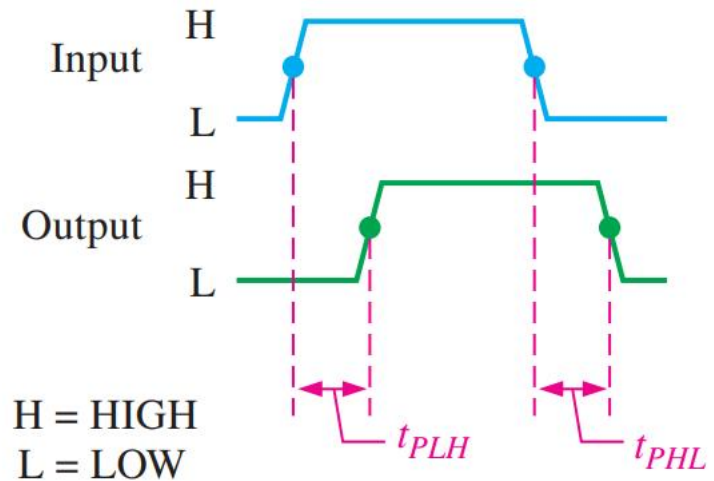


FIGURE 15-9 Propagation delay times.

